DATA QUALITY / MARKETING OPERATIONS

Marketing Data Health Monitor

A data quality monitoring application for identifying, prioritizing, and investigating issues in B2B marketing records.

Portfolio project | Python • SQL Server • Streamlit • Plotly

Synthetic dataset — independently developed

Made with GAMMA

01 / BUSINESS PROBLEM

Marketing decisions depend on trustworthy data

Incomplete, duplicated, or inconsistent records can undermine everyday marketing operations.

DATA COMPLETENESS — Missing fields: Key contact and campaign attributes may be absent.

DATA VALIDITY — Invalid values: Malformed email addresses can affect outreach readiness.

DATA CONSISTENCY — Duplicate / variant records: Repeated contacts and inconsistent company names complicate segmentation.

Project objective: Build a practical workflow that surfaces data quality issues, makes them measurable, and helps an operator decide which records need attention first.

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A controlled dataset with realistic quality issues

Synthetic B2B marketing lead data was generated for demonstration; it does not contain SAP or customer data.

TOTAL RECORDS:

2,050

— 2,000 base records + 50 duplicate-row copies

ATTRIBUTES:

14

— Contact, company, campaign, region, lifecycle and consent fields

FLAGGED RECORDS:

410

— Records identified for review

AVERAGE SCORE:

95.72

— Illustrative project scoring logic

Issues intentionally introduced: missing values, invalid emails, duplicate records, and company-name variants.

The dataset supports repeatable testing of validation logic and dashboard behavior.

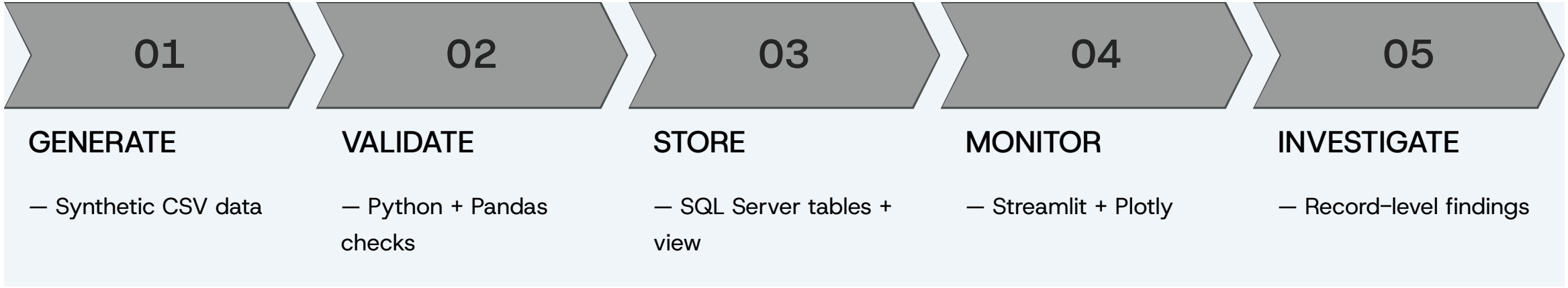
Quality scores and severity labels are project-defined indicators, not external or SAP standards.

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A repeatable pipeline from raw records to review

Implemented in the application

The workflow separates source data, validation logic, database storage, and the operator-facing interface.



Design principle: Keep validation rules explicit and queryable, then present the results in a workflow that supports both portfolio-level monitoring and record-level investigation.

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Monitor the full dataset and explore individual records

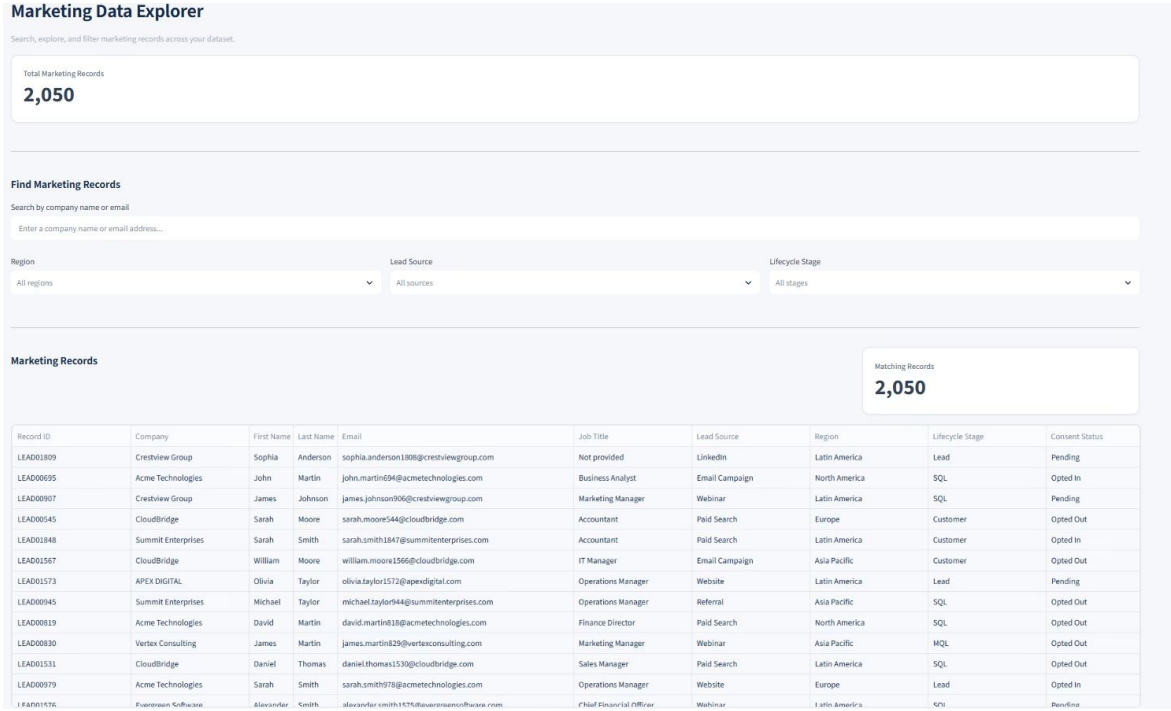
Implemented in the application

The overview highlights data health; the explorer lets an operator search and filter the underlying records.

OVERVIEW DASHBOARD



DATA EXPLORER



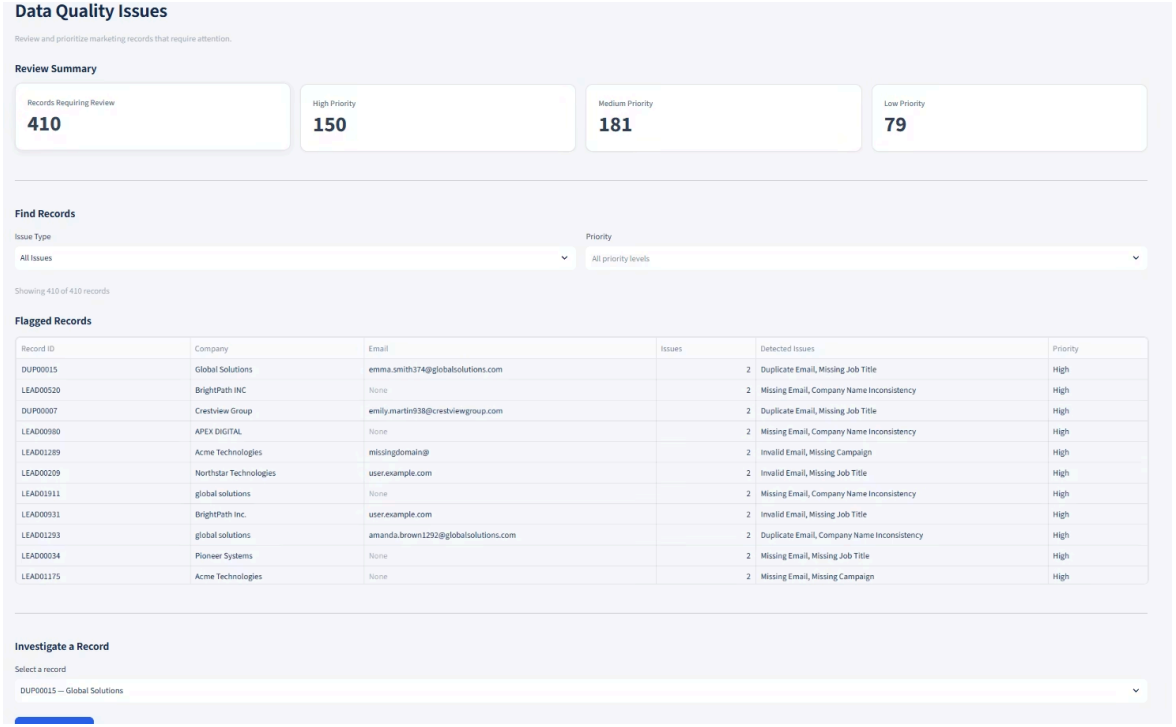
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Prioritize issues and investigate individual records

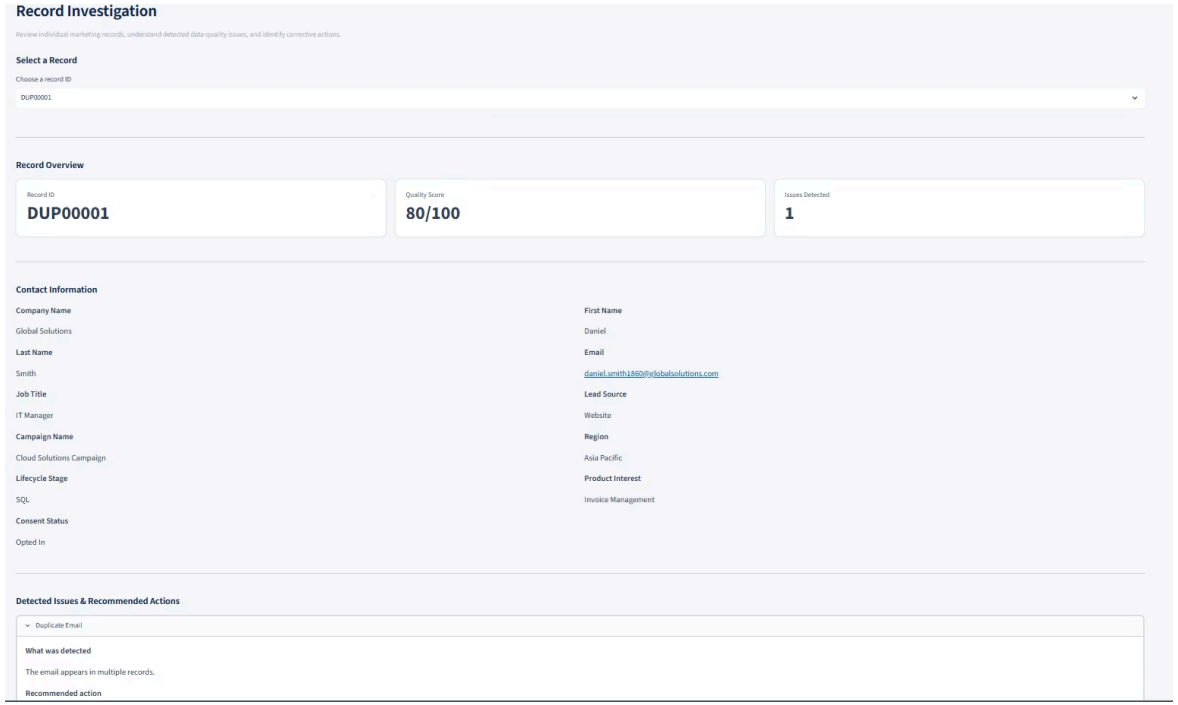
Implemented in the application

The review queue surfaces records requiring attention; investigation explains the detected issue and suggests a next action.

DATA QUALITY ISSUES



RECORD INVESTIGATION



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The monitor identified 410 records requiring review

Implemented in the application

Counts below reflect the current synthetic dataset and the implemented validation rules.

MISSING EMAIL:

62

— 3.02% of all records

INVALID EMAIL:

41

— Flagged invalid email records

DUPLICATE EMAIL:

47

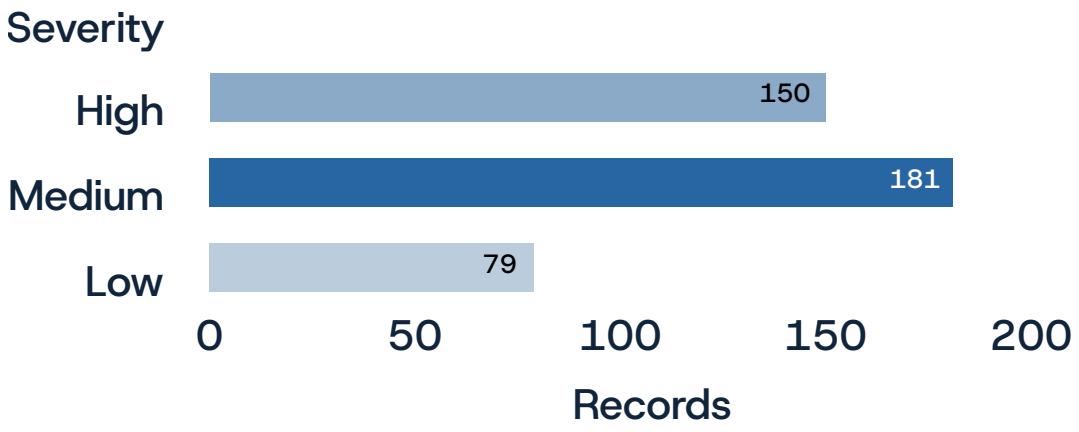
— Valid duplicate email records

COMPANY NAME INCONSISTENCIES:

77

— Rows flagged as inconsistent

Review severity (record-level)



Severity is assigned using the highest-priority issue on each flagged record:

- High — email and duplicate issues
- Medium — company and campaign issues
- Low — missing job title

Severity definitions are illustrative project rules.

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Turn detection into consistent data stewardship

Proposed recommendation — not implemented

These are recommendations for a future operating model—not policies implemented in the application or statements of SAP policy.

RETENTION — Define retention periods by data purpose and record type; review inactive records on a scheduled cadence.

DEDUPLICATION — Establish matching keys, normalization rules, and a review path before merging potential duplicates.

CONSENT — Make consent status visible and define ownership for validating and updating preference records.

MASS DELETION — Use approval, scope validation, audit logging, and post-action checks for bulk deletion workflows.

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Make data quality actionable—not just measurable

Proposed recommendation — not implemented

A lightweight operating rhythm can help teams move from issue visibility to resolution.

01 Assign an owner and target resolution path for each high-priority issue category.

02 Review quality trends regularly and investigate recurring sources of defects.

03 Document definitions for key fields, validation rules, and severity levels.

04 Communicate findings in plain language: what is affected, why it matters, and what action is needed.

05 Track issue resolution over time to understand whether data quality is improving.

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Reliable records support more confident marketing operations

Potential value

The project demonstrates a workflow that could help teams prioritize data hygiene and communicate quality risks.

VISIBILITY — Shared view: A consistent picture of data health across the dataset.

PRIORITIZATION — Focused review: Issue categories and severity help direct attention.

TRACEABILITY — Record context: Operators can inspect why a record was flagged.

Potential downstream relevance: Cleaner, more consistent marketing data can support segmentation, campaign execution, customer engagement, and more trusted reporting.

Built with Python, SQL Server, Streamlit & Plotly.

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